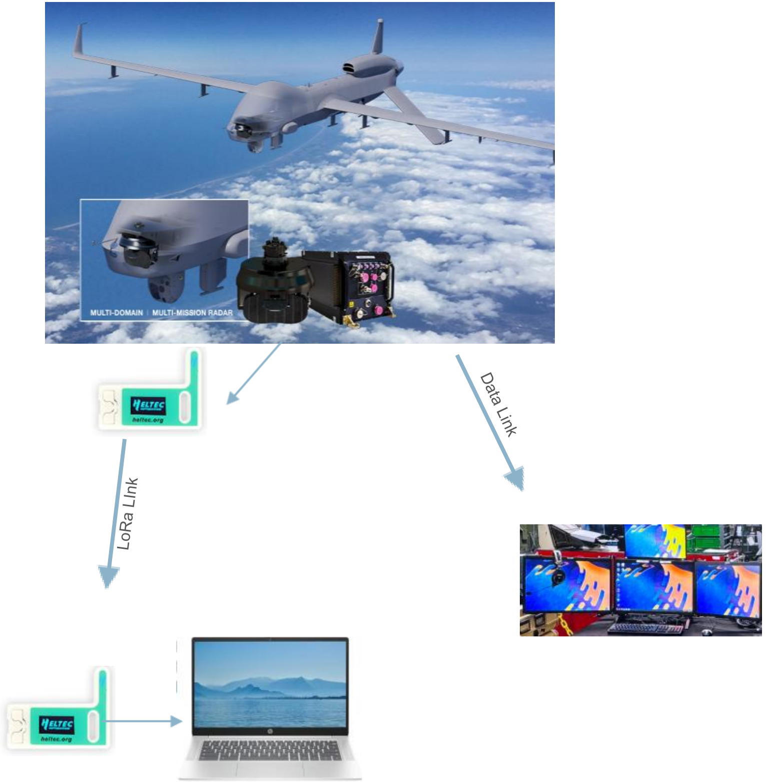
Project Overview

WES207 SP26

Project Idea

Goal: Develop a LoRa telemetry system that simulates radar behavior and wirelessly transmits operational status to a ground station.

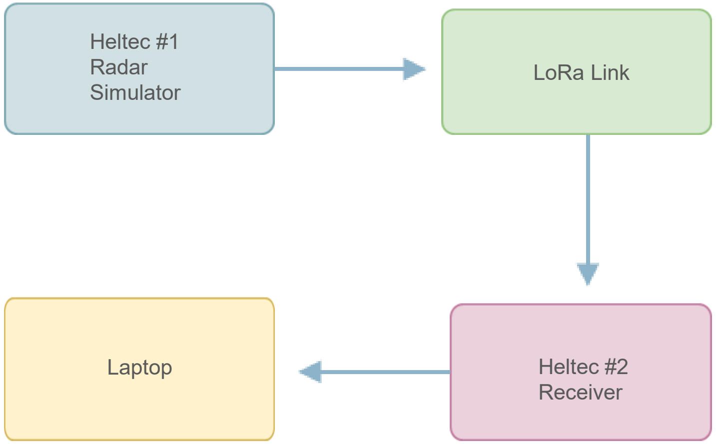
- Monitors mode, boot time, run time, sleep/awake state, and health status
- Displays live telemetry on a laptop dashboard for real-time monitoring
- Demonstrates remote monitoring when direct access to radar hardware is limited



System Overview

Architecture:

- Heltec #1 simulates radar behavior
    - Generates mode changes
    - Tracks boot time, run time, and sleep/awake state
    - Sends health status
    - LoRa link transmits telemetry packets
- Heltec Board #2 receives packets
- Laptop dashboard displays live radar status



Minimal Viable Product (Week 8)

By Week 8 the system will:

- Simulate radar mode transmissions
- Track and transmit boot-up time
- Track and transmit run time
- Track and transmit sleep/awake state
- Track and transmit health status
- Send telemetry packets over LoRa every 1–5 seconds
- Receive packets reliably
- Display live telemetry on a simple laptop dashboard
- Complete a short-range system test

Weekly Milestones

Weeks	Milestone	Action Item
1 (Apr 3–9)	Requirements	Define telemetry fields, confirm MVP, outline radar states
2 (Apr 10–16)	Architecture	Create system diagram, data packet format, development environment setup
3 (Apr 17–23)	LoRa Link	Send/receive simple packets, validate RSSI/SNR
4 (Apr 24–30)	Radar Simulation	Implement mode changes, boot/run timers, sleep/awake logic
5 (May 1–7)	Ground Station	Receive packets, forward via serial, attach metadata
6 (May 8–14)	Dashboard	Display mode, timers, health, basic UI
7 (May 15–21)	Data Logging	Packet loss counter, CSV logging, error flags
8 (May 22–28)	MVP Demo	Outdoor test, full end-to-end demonstration

Bi-Weekly Sprints

Sprint	Dates	Focus	Tasks
Sprint 1	Apr 3 – Apr 16	Requirements and Architecture	Define telemetry fields, confirm MVP, create system diagram, choose data display format, set up development environment
Sprint 2	Apr 17 – Apr 30	LoRa Link and Radar Simulation	Establish LoRa communication, implement radar mode simulation, implement boot time and run time counters, implement sleep/awake logic
Sprint 3	May 1 – May 14	Ground Station and Dashboard	Build receiver firmware, forward packets to laptop, display mode and timers, display sleep/awake state, display health status
Sprint 4	May 15 – May 28	Data Logging and MVP Testing	Implement packet loss tracking, implement CSV logging, add error and warning flags, perform outdoor test, demonstrate full MVP
Sprint 5	May 29 – June 5	Final Deliverables	Complete final report, complete final presentation, polish final demo

Extensions After Minimal Viable Product

- **Drone-Mounted LoRa Telemetry Pod (Mobile Link Testing)**
  - Attach the Heltec board to an off-the-shelf drone as an independent telemetry module.

- **Power BI Telemetry Analytics Dashboard**
  - Import logged LoRa data into Power BI for professional-grade visualization.

- **Engineering Brief / Technical Paper**
  - Document the system architecture, link performance, and test results in a short engineering paper.

Risks

- **Risk 1 – Cannot use actual radar hardware (Proprietary System)**
    - Impact: Real radar telemetry cannot be accessed or tested.
    - Mitigation: Simulate radar behavior (mode, boot time, run time, sleep/awake state).
- **Risk 2 – Packet loss during motion or obstruction**
    - Impact: A moving platform can cause packet loss.
    - Mitigation: Add packet counters, retries, and logging to detect loss.